Health Indicators Research: Phase 1

NASA Space Apps 2026 | Health Monitoring Software | Prepared by Eman Saleem (Research) for Abdul Rafay | 11 November 2026 | **Status: draft, sources being verified**

What this covers (from the Team Work Distribution plan):

- **Part 1**: which health indicators matter in space and why, with normal / warning / urgent ranges and sources.
- **Part 2**: which measurements can be done in space, with what equipment, why each is needed, and a short "how to measure" instruction.
- **Part 3**: NASA Human Research Program material and other agency sources (ESA, JAXA, CSA).

The same table is also available as **indicators_for_developers.csv** for direct use in code.

Please note:

- Cells shaded light yellow in the Basis column contain a **TEAM CHOICE** value: our design decision, not a medical standard. Please approve or change.
- **Decision needed:** keep fasting blood sugar? Astronauts do not test it daily.
- Green = normal, Yellow = warning / watch, Red = urgent / contact flight surgeon. The app is decision support, not a medical diagnosis.

Part 1: Indicators and ranges

Indicator	Normal	Warning	Urgent	Basis for the numbers	Why it matters in space	Sources
Resting heart rate beats/min	51 - 90	41 - 50 or 91 - 130	40 or less or 131 or more	NEWS2 bands (score 0 = green, 1-2 = yellow, 3 = red). MedlinePlus normal resting = 60-100.	Weightlessness changes blood circulation and how the heart works.	Royal College of Physicians NASA HRP
Blood pressure - systolic (top number) mmHg	111 - 119	91 - 110 or 120 - 180	90 or less or more than 180	Low side: NEWS2 (<=90 = score 3). High side: American Heart Association (120+ elevated, >180 = hypertensive crisis).	After spaceflight the body struggles to hold blood pressure on standing.	American Heart Association Royal College of Physicians
Blood pressure - diastolic (bottom number) mmHg	60 - 79	80 - 120	more than 120	American Heart Association (80+ = stage 1, >120 = crisis). MedlinePlus normal from 60.	Same as systolic.	American Heart Association NIH MedlinePlus
Breathing rate breaths/min	12 - 20	9 - 11 or 21 - 24	8 or less or 25 or more	NEWS2 bands. MedlinePlus normal resting = 12-18.	Early sign of low oxygen or high cabin CO2.	Royal College of Physicians NASA HRP
Blood oxygen (SpO2), on air %	96 or more	92 - 95	91 or less	NEWS2 SpO2 scale 1 (>=96 = 0, 94-95 = 1, 92-93 = 2, <=91 = 3).	Spacewalk preparation can raise the risk of mild hypoxia.	Royal College of Physicians NASA HRP
Body temperature deg C	36.1 - 38.0	35.1 - 36.0 or 38.1 - 39.0	35.0 or less or 39.1 or more	NEWS2 bands (39.1+ scores 2 in NEWS2; team marks it red). MedlinePlus normal = 36.5-37.3.	The immune system weakens in space; fever can be the first infection sign.	Royal College of Physicians NASA HRP

Fasting blood sugar mg/dL	70 - 99	54 - 69 or 100 - 125	below 54 or 126 or more	CDC: normal <=99, prediabetes 100-125, diabetes >=126; low <70, severe low <54. Red here = contact flight surgeon, not always an emergency.	Diet and activity change on long missions. Note: not a routine daily astronaut test; team to decide whether to keep it.	CDC CDC
Sleep last night hours	7 or more	6 - 6.9	less than 6 for 3 nights in a row	CDC: healthy adult sleep = 7+ h. NASA schedules 8.5 h; astronauts averaged ~6 h (NASA-funded Lancet study). The 3-night red rule is a TEAM CHOICE, not a medical standard.	Light-dark cycle is broken (16 sunrises a day on ISS); poor sleep hurts performance.	CDC NASA HRP
Cabin CO2 (1-hour average) mmHg	3.0 or less	3.1 - 4.5	more than 4.5	NASA-STD-3001 Volume 2 [V2 6004] limit = 3 mmHg (1-hr avg). OCHMO TB-004: headaches reported 2.8-4.5 mmHg; fatigue, poor sleep, nausea above 4.5.	CO2 builds up inside a closed spacecraft.	NASA OCHMO NASA OCHMO
Radiation dose - career total mSv	less than 300	300 - 599	600 or more	NASA limit: career < 600 mSv (red). 300 mSv (= 50% of limit) yellow line is a TEAM CHOICE. Separate alert: one solar particle event limited to 250 mSv.	Space radiation raises cancer risk; outside Earth's magnetic field it is much higher.	NASA OCHMO NASA Human Research Program
Exercise today minutes	120 or more	60 - 119	less than 60 for 2 days in a row	JAXA: ISS crew exercise 2 hours a day. Yellow/red cut-offs are a TEAM CHOICE.	Bones lose about 1-2% mass per month in microgravity; exercise slows it.	JAXA NASA NTRS
Mood / stress self-check score 1-5 (5 = very good)	4 - 5	3	1 - 2 for 2 days in a row	TEAM CHOICE scale (not a clinical test). Linked to NASA behavioral health risk.	Isolation and distance from Earth affect mental health.	NASA HRP NASA Human Research Program

NEWS2 = National Early Warning Score 2, the standard UK hospital tool for spotting a patient getting worse. Extra radiation alert: a single solar particle event is limited to 250 mSv (NASA OCHMO TB-020).

Part 2: Measurements in space

"How often" is our suggested routine for the app (team choice), not a NASA schedule.

Indicator	Equipment in space	Why this exam is needed	How to measure (text for the app)	How often	Source
Resting heart rate	Heart rate monitor; CSA Bio-Monitor shirt	Microgravity changes how the heart and blood vessels work.	Sit still 5 min after waking. Count pulse at wrist for 30 s and x2, or read the heart rate monitor.	Daily, morning	NASA NTRS
Blood pressure	Blood pressure cuff (ISS Medical Diagnostics Pack, CHeCS)	Fluid shifts in space; on return, blood pressure can drop on standing.	Sit, rest 5 min, arm at heart level, cuff on bare upper arm. Take 2 readings 1 min apart and use the average.	Daily, morning	NASA NTRS
Breathing rate	Timer, or Bio-Monitor shirt	Early sign of low oxygen or high cabin CO2.	Count breaths for 30 s while relaxed and x2. Don't change your breathing while counting.	Daily, morning	Canadian Space Agency (CSA)
Blood oxygen (SpO2)	Fingertip pulse oximeter; Bio-Monitor shirt	Detects low oxygen (hypoxia), e.g. around spacewalks.	Clip pulse oximeter on a warm finger, stay still 30-60 s, read the steady number.	Daily + before/after spacewalk	Canadian Space Agency (CSA)
Body temperature	Thermometer	Immune system weakens in space; fever = possible infection.	Measure under the tongue, mouth closed, no drink for 15 min before.	Daily, and when feeling unwell	NIH MedlinePlus
Fasting blood sugar	Glucose meter	Diet and activity change on long missions.	Before breakfast after 8+ h without food. Finger prick, glucose meter.	Weekly or as planned by team	CDC
Sleep	Actigraphy watch (used in NASA Standard Measures)	16 sunrises a day on ISS disturb sleep; poor sleep hurts performance.	Enter hours slept. On ISS a wrist actigraphy watch records sleep.	Daily, morning	NASA HRP
Cabin CO2	Vehicle CO2 sensor	CO2 builds up in a closed cabin and causes headaches.	Read from spacecraft environment sensor (not a body measurement). Watch for headache.	Continuous (app shows daily average)	NASA OCHMO
Radiation dose	Personal dosimeter	Space radiation raises cancer risk; NASA sets a career limit.	Read personal dosimeter; app adds today's dose to the running total.	Daily	NASA OCHMO
Exercise	Exercise devices log	Without gravity bones lose 1-2% mass per month; exercise slows it.	Enter minutes on treadmill, bike and resistance device.	Daily	JAXA
Mood / stress	App question	Isolation and confinement affect mental health.	Answer: 'How do you feel today?' 1-5. Optional note.	Daily, evening	NASA HRP

Part 3: Sources

NASA Human Research Program (HRP)

#	Source	Used for	Link
1	**NASA Human Research Program**: 5 Hazards of Human Spaceflight	Radiation, isolation, distance, gravity, closed environment	https://www.nasa.gov/hrp/hazards/
2	**NASA Human Research Program**: Risks of Human Spaceflight	List of all HRP risks with links	https://www.nasa.gov/hrp/risks/
3	**NASA HRP**: Risk of Cardiovascular Adaptations	Why heart rate and blood pressure matter in space	https://www.nasa.gov/reference/risk-of-cardiovascular-adaptations/
4	**NASA HRP**: Risk of Altered Immune System Responses	Why temperature / infection signs matter	https://www.nasa.gov/reference/risk-of-altered-immune-system-responses/
5	**NASA HRP**: Risk of Hypoxia	Why SpO2 and breathing rate matter	https://www.nasa.gov/reference/risk-of-hypoxia/
6	**NASA HRP**: Risk from Inadequate Sleep and Irregular Schedules	Sleep risk; 16 sunrises/sunsets per day on ISS	https://www.nasa.gov/reference/risk-from-inadequate-sleep-and-irregular-schedules/
7	**NASA HRP**: Risk of Spaceflight-Induced Bone Changes	Why exercise and bone matter	https://www.nasa.gov/reference/risk-of-spaceflight-induced-bone-changes/
8	**NASA HRP**: Risk of Behavioral Conditions and Psychiatric Disorders	Why mood / stress check-in matters	https://www.nasa.gov/reference/risk-of-behavioral-conditions-and-psychiatric-disorders/
9	**NASA HRP**: Risk of Inadequate Food and Nutrition	Why diet / blood sugar can matter	https://www.nasa.gov/reference/risk-of-inadequate-food-and-nutrition/
10	**NASA HRP**: Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)	Why headache / vision change symptoms matter	https://www.nasa.gov/reference/risk-of-spaceflight-associated-neuro-ocular-syndrome-sans/
11	**NASA HRP**: Spaceflight Standard Measures	What NASA measures on every ISS crew member	https://www.nasa.gov/humans-in-space/spaceflight-standard-measures-characterizing-how-humans-adapt-in-space
12	**NASA**: NASA's Twins Study Results Published	1-year mission story for the presentation	https://www.nasa.gov/humans-in-space/nasas-twins-study-results-published-in-science-journal/

NASA standards and technical reports

#	Source	Used for	Link
13	**NASA OCHMO**: NASA-STD-3001 Volume 1: Crew Health	NASA's official crew health standard	https://standards.nasa.gov/standard/NASA/NASA-STD-3001_VOL_1
14	**NASA OCHMO**: OCHMO Human Spaceflight Standards + Technical Briefs	Library of short technical briefs	https://www.nasa.gov/ochmo/health-operations-and-oversight/hsa-standards/
15	**NASA OCHMO**: Technical Brief OCHMO-TB-004: Carbon Dioxide	CO2 limit 3 mmHg (1-hr avg); symptoms at 2.8-4.5 and >4.5 mmHg	https://www.nasa.gov/ochmo-tb-004-carbon-dioxide-2/
16	**NASA OCHMO**: Technical Brief OCHMO-TB-020: Design for Ionizing Radiation Protection	Career limit <600 mSv; solar particle event limit 250 mSv	https://www.nasa.gov/wp-content/uploads/2023/03/radiation-protection-technical-brief-ochmo.pdf

17	**NASA NTRS**: Crew Health Care System (CHeCS) report	ISS medical kit includes blood pressure cuffs (Medical Diagnostics Pack)	https://ntrs.nasa.gov/api/citations/20110011351/downloads/20110011351.pdf
18	**NASA NTRS**: Digital Astronaut Project Bone Remodeling Model (1-2% bone loss/month)	Bone mass loss 1-2% per month in microgravity	https://ntrs.nasa.gov/citations/20140006747
19	**NASA**: NASA Open Science Data Repository (OSDR)	Public space biology data (mostly non-human)	https://osdr.nasa.gov

Other space agencies (ESA, JAXA, CSA)

#	Source	Used for	Link
20	**European Space Agency (ESA)**: Space medicine	ESA overview of astronaut health and space medicine	https://www.esa.int/About_Us/Welcome_to_ESA/Space_medicine
21	**JAXA**: Bone loss research (health benefits)	Bone loss in space ~10x rate of osteoporosis	https://iss.jaxa.jp/en/iss/benefits/health/2boneloss
22	**JAXA**: Bisphosphonates as a Countermeasure to Bone Loss	ISS crew exercise 2 hours a day	https://iss.jaxa.jp/en/kiboexp/theme/first/bisphosphonates/
23	**Canadian Space Agency (CSA)**: Bio-Monitor: Keeping an eye on astronauts' vital signs	Wearable smart shirt that records vital signs on the ISS	https://asc-csa.gc.ca/eng/sciences/bio-monitor.asp

Medical reference ranges

#	Source	Used for	Link
24	**NIH MedlinePlus**: Vital signs (Medical Encyclopedia)	Normal adult resting ranges: BP, breathing, pulse, temperature	https://medlineplus.gov/ency/article/002341.htm
25	**Royal College of Physicians**: NEWS2 Chart 1: The NEWS scoring system	Clinical warning bands for HR, breathing, SpO2, BP, temperature	https://www.rcp.ac.uk/media/alxev00t/news2-chart-1_the-news-scoring-system_0_0.pdf
26	**American Heart Association**: Understanding Blood Pressure Readings	BP categories: normal <120/80 ... crisis >180/120	https://www.heart.org/bplevels
27	**CDC**: Diabetes Testing	Fasting sugar: normal <=99, prediabetes 100-125, diabetes >=126 mg/dL	https://www.cdc.gov/diabetes/diabetes-testing/index.html
28	**CDC**: Low Blood Sugar (Hypoglycemia)	Low <70 mg/dL; severe <54 mg/dL	https://www.cdc.gov/diabetes/about/low-blood-sugar-hypoglycemia.html
29	**CDC**: 1 in 3 adults don't get enough sleep (healthy = 7+ hours)	Healthy adult sleep = 7 or more hours	https://archive.cdc.gov/www_cdc_gov/media/releases/2016/p0215-enough-sleep.html
30	**Lancet Neurology**: Barger et al. 2014, Sleep deficiency in astronauts (Lancet Neurology), NASA-funded	Astronauts averaged ~6 h sleep vs 8.5 h scheduled	https://doi.org/10.1016/S1474-4422(14)70122-X

Space Apps challenge and rules

#	Source	Used for	Link
31	**NASA Space Apps**: Space Apps 2026 challenge: Create Health Monitoring Software for Astronauts on Space Missions	Official challenge + its Resources list (judges' own links). Open this first and copy every resource link.	https://www.spaceappschallenge.org/2026/challenges/create-health-monitoring-software-for-astronauts-on-space-missions/

32	**NASA Space Apps**: 2026 NASA Space Apps Challenge (timeline)	Dates: hackathon Nov 14-15, submission closes Nov 15 11:59 PM local	https://www.spaceappschallenge.org/2026/
33	**NASA Space Apps**: Participant FAQs / Important Dates	Nov 13 = Global Offers + Submission and Judging guides released	https://www.spaceappschallenge.org/resources/participant-faqs/
34	**NASA Space Apps**: Judging & Awards Guide	5 judging criteria; References field rule; English rule	https://spaceappschallenge.org/resources/judging-awards-guide
35	**NASA Space Apps**: Project Submission Guide	Demo format (last year: 30 s video or 7 slides). Re-check 2026 version on Nov 13	https://www.spaceappschallenge.org/resources/project-submission-guide/